

Program : Diploma in Renewable Energy	
Course Code : 6292A	Course Title: Conventional and Non Conventional Energy Sources
Semester : 6	Credits: 4
Course Category: Open Elective Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the concepts of conventional energy sources
- To impart the basic knowledge of solar and wind energy
- To understand the concepts of tidal and ocean energy
- To understand the concepts of biomass energy, small hydro sources and geothermal energy sources

Course Prerequisites:

Topic	Course Code	Course name	Semester
Different renewable energy resources	2001	Environmental Science	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basic concepts of conventional energy sources and energy storage	13	Understanding
CO2	Illustrate the solar and wind energy conversion system	16	Understanding
CO3	Explain the production of energy from oceans and tides	13	Understanding
CO4	Summarize the concepts of biomass, small hydro and geothermal energy sources.	16	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2				3		
CO2	2				3		
CO3	2				3		
CO4	2				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basic concepts of conventional energy sources and energy storage.		
M1.01	Summarize the classification of Energy resources.	3	Understanding
M1.02	Illustrate the concepts of Steam Power Station.	3	Understanding
M1.03	Elaborate the concepts of Hydro and Nuclear power plants.	3	Understanding
M1.04	Explain different types of Energy storage methods.	4	Understanding
Contents: Conventional Energy Resources: Introduction-classification- Conventional Energy Resources -availability and their limitations- Non-Conventional Energy Resources – classification-advantages-limitations. Steam power station- Schematic arrangement-choice of site-efficiency-equipment of steam power station-advantages-limitations Hydro electric power station- Schematic arrangement-choice of site-constituents of Hydro electric plant -advantages-limitations Nuclear power stations- Schematic arrangement-choice of site-advantages-limitation. Energy storage- Necessity of energy storage-methods- Mechanical,Electrochemical, Thermal energy storage.			

CO2	Illustrate the solar and wind energy conversion systems		
M2.01	Outline the concepts of solar energy.	4	Understanding
M2.02	Illustrate the solar thermal systems.	4	Understanding
M2.03	Explain the concept of solar photovoltaic systems.	4	Understanding
M2.04	Summarize the wind energy conversion system.	4	Understanding
	Series Test – I	1	

Contents :

Solar energy: Scope and potential of solar energy-advantages and disadvantages of solar energy -applications of solar energy(listing only)-basic concepts of solar energy conversion techniques (solar thermodynamics, solar photovoltaic and greenhouse effect)-Advantages and Disadvantages.

Solar Thermal Systems: Principle-Solar thermal collectors-classification based on temperature (low, medium and high temperature)-Types of solar collectors-Concentrated and Non-Concentrated- Applications-Basic operation of Solar Cooker,Solar Water Heater,Solar Distillation-Advantages and Disadvantages.

Solar PV Systems:Solar Photovoltaic-solar cell fundamentals-classification-construction.

Classification of solar PV system- Standalone and grid connected (Block diagram)-Advantages and Disadvantages

Wind Energy:Wind-properties-Factors affecting distribution of wind energy-advantages-disadvantages-Site selection considerations-basic principles of Wind Energy Conversion Systems (WECS)-basic components of a WECS-Advantages and Disadvantages.

CO3	Explain the production of energy from oceans and tides.		
M3.01	Explain the OTEC principle and methods of OTEC power generation.	4	Understanding
M3.02	Illustrate the production of energy from waves.	3	Understanding
M3.03	Summarize the principle and components of a tidal power plant.	3	Understanding
M3.04	Outline the classification of tidal power plants.	3	Understanding

Contents :

Ocean Thermal Energy:Introduction-OTEC principle-Site selection criteria-advantages and limitations-Types-Open cycle(Claude cycle)-Closed cycle(Anderson cycle)-Hybrid cycle (Block diagram description).

Energy from Ocean Waves:Introduction-Energy and power from waves- Advantages and limitations of wave energy-Wave energy conversion devices-Floats,High level Reservoir Wave machine,Dolphin Type Wave power Machine.

Tidal Energy:Introduction-Origin and nature of Tidal energy-Basic principle of tidal power-components of tidal power plants.

Classification of Tidal power plants:Single basin arrangement-Classification-Single ebb cycle-Double ebb cycle-Double basin arrangement-Advantages and limitations.

CO4	Summarize the concepts of biomass,small hydro and geothermal energy sources.	16	Understanding
M4.01	Explain the concepts of biomass energy and biomass conversion technologies.	4	Understanding
M4.02	Summarize the concepts of small hydro resources.	4	Understanding
M4.03	Outline geothermal energy sources.	4	Understanding
M4.04	Illustrate the working of fuel cells.	4	Understanding
	Series Test – II	1	

Contents :

Biomass Energy-Introduction-Biomass resources-Bio chemical conversion process - Anaerobic digestion,Fermentation-advantages,disadvantages.

Biogas-Biogas production from Biomass-Classification of Biogas plants-Continuous and Batch types,The Dome and the Drum types-Factors affecting generation of Biogas-site selection criteria.

Small Hydro Resources-Introduction-nature of small hydro development-classification of SHP, advantages and disadvantages.

Geothermal Energy-Introduction-Types of geothermal sources (listing only)-classification of Geothermal power plants- Liquid dominated,Vapor dominated-Advantages and limitations-applications.

Fuel Cell- Introduction-Working Principle-Types (listing only)-Working of Hydrogen Fuel Cell(Alkaline, Phosphoric acid,Proton Exchange membrane Fuel cell)-Advantages and disadvantages.

Text / Reference

T/R	Book Title/Author
T1	G.D Rai - Non conventional Energy Sources, Khanna Publishers
T2	Khan B H-Non conventional Energy Resources, Tata McGraw Hill, 2009
T3	R K Rajput- Non-conventional energy sources and utilization,,S chand Pvt Limited
T4	Anand Temdulkar- Non-conventional energy sources- S K Kataria
R1	Sawhney .G.S Non Conventional Energy Resources, PHI Learning, 2012

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103103206
2	https://openei.org/wiki/Wind_energy
3	https://www.nrel.gov/research/re-biomass.html
4	https://www.yourelectricalguide.com/2018/05/tidal-power-plant-working-principle.html
5	https://otecorporation.com/otec
6	https://www.irena.org/geothermal